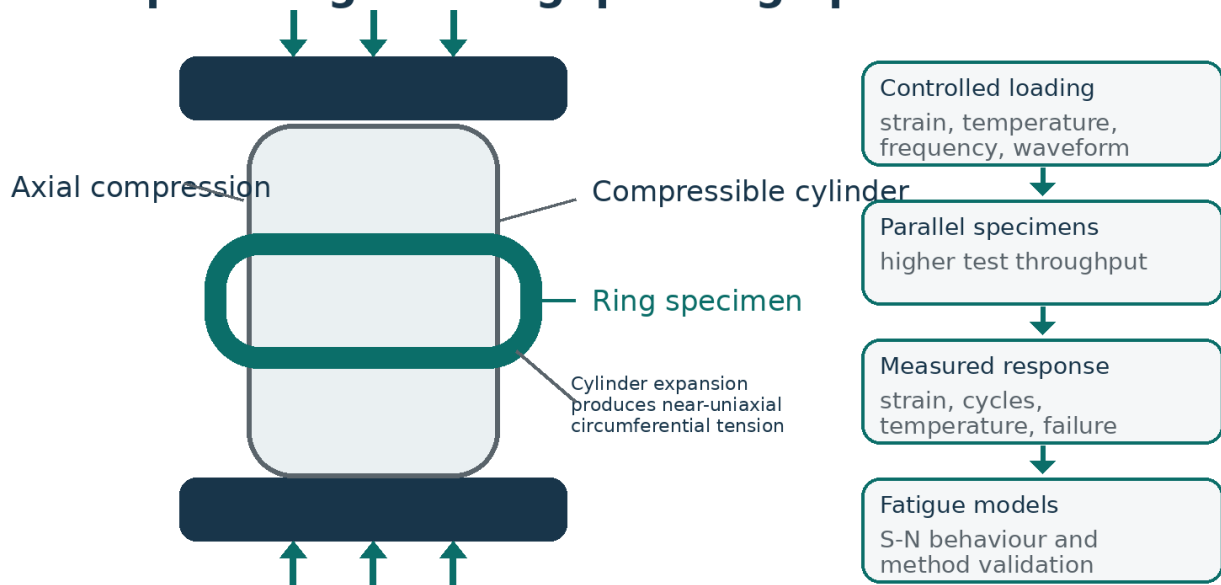


SELECTED ENGINEERING AND RESEARCH PROJECTS

Mo Balfour-Lochner

PhD candidate, University of Canterbury

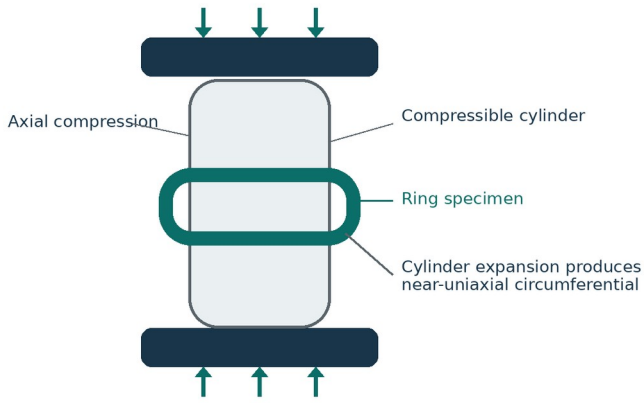
Conceptual high-throughput ring-specimen test



I work across experiments, software and hardware. This short portfolio selects the projects most relevant to an applied emerging-technology internship: fatigue research, AI/ML, embedded systems, computational modelling and engineering design taken through to manufacture.

Experimental mechanics	Python and AI/ML	Embedded systems	Computational modelling	Design and manufacture
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1. PHD RESEARCH - HIGH-THROUGHPUT FATIGUE TESTING



Research question

Can a ring-specimen method increase fatigue-test throughput while retaining controlled, interpretable loading and useful strain measurements?

Current programme

- Developing and validating the method on a purpose-built 500 kN machine.
- Initial material focus: 5000- and 6000-series aluminium.
- Closed-loop strain and temperature control, higher-frequency loading and variable-amplitude capability.
- Current design work includes specimen interactions, instrumentation and lower-cost strain-measurement options.

Why it is relevant

- Experimental system design and commissioning
- Sensor selection and data quality
- Mechanics-based interpretation
- Large test datasets and automation
- Translation of research into a usable test method

Conference presentation

Presented "High-Cycle Fatigue Characterization via Novel High Throughput Experimentation" at the 2026 Society for Experimental Mechanics Annual Conference, Norfolk, Virginia.

Related computational work

Building density-functional-theory capability with VASP 5.4.4. I have completed the unstrained relaxation/static stage for monolayer MoS₂ and am progressing to strain-dependent band-structure calculations.

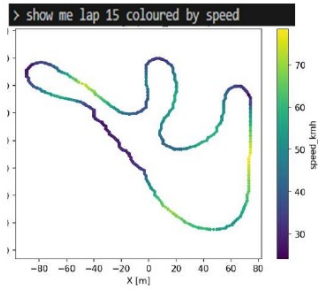
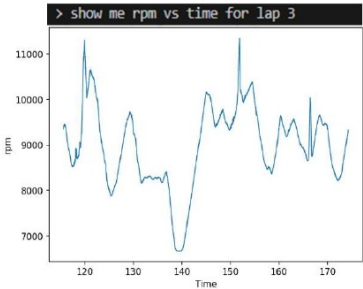
2. AI, MACHINE LEARNING AND LOCAL DATA TOOLS

Offline motorsport telemetry assistant

A local Python/Ollama prototype for asking natural-language questions about MoTeC telemetry logs. It identifies headers and channels, computes derived values and returns plots or track views without sending team data to a cloud service. The current development work is deliberately practical: improving channel matching, lap segmentation and behaviour when logs are incomplete.

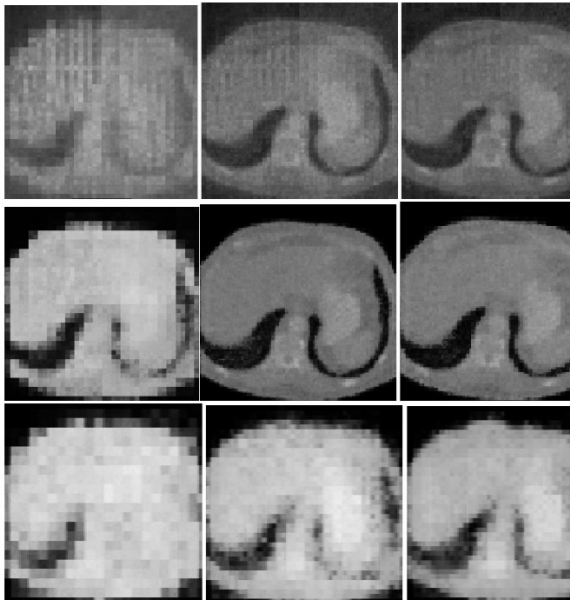
```
> which lap had the highest speed?
TOP1 laps by max speed_kmh:
lap
21 83.88
> what is our optimal lap time?
Theoretical best (sector_count=10): 43.382
sector lap dt
0 21 4.19
1 22 3.81
2 23 4.21
3 21 6.78
4 23 3.84
5 21 3.39
6 22 3.80
7 21 4.38
8 20 4.21
9 21 4.78
```

Local natural-language queries mapped to telemetry calculations and visual outputs



2B. NEURAL ATTENUATION FIELD FOR 3D X-RAY RECONSTRUCTION

The network represents a continuous attenuation field rather than a fixed voxel grid. Known cone-beam geometry is used to cast and sample rays; the model predicts attenuation at each point and a discretised Beer-Lambert integral generates a synthetic projection for training.



```
# model_attn.py
1 import tensorflow as tf
2
3
4 class ResBlock(tf.keras.layers.Layer):
5     def __init__(self, hidden):
6         super().__init__()
7         self.d1 = tf.keras.layers.Dense(hidden, activation="linear")
8         self.act = tf.keras.layers.LeakyRelu(0.2)
9         self.d2 = tf.keras.layers.Dense(hidden, activation="linear")
10
11     def call(self, x):
12         h = self.act(self.d1(x))
13         h = self.d2(h)
14         return x + self.act(h) # residual
15
16
17 class ModelAttnDot(tf.keras.layers.Layer):
18     def __init__(self, encoder,
19                  hidden=64,
20                  num_blocks=3,
21                  n_points=3072):
22         super().__init__()
23         self.encoder = encoder
24         self.n_points = n_points
25         self.hidden = hidden
26
27         self.fc_in = tf.keras.layers.Dense(hidden)
28
29         # attention block
30         self.in_attn = tf.keras.layers.LayerNormalization(epsilon=1e-5)
31         self.attn = tf.keras.layers.MultiHeadAttention(
32             num_heads=2, key_dim=hidden // 2)
33
34         self.blocks = [ResBlock(hidden) for _ in range(num_blocks)]
35
36         self.fc_out = tf.keras.layers.Dense(1, activation=tf.nn.softplus)
37
38     def call(self, pts):
39         """
40         pts: [N, 3]
41         """
42         x = self.encoder(pts)
43         x = self.fc_in(x)
44
45         N = tf.shape(x)[0]
46         if tf.math.equal(tf.math.floormod(N, self.n_points), 0):
47             n = N // self.n_points
48             x3 = tf.reshape(x, (n, self.n_points, self.hidden))
49
50             attn_out = self.attn(self.in_attn(x3), self.in_attn(x3))
51             x3 = x3 + attn_out
52             x = tf.reshape(x3, (-1, self.hidden))
53
54         for blk in self.blocks:
55             x = blk(x)
56
57         return self.fc_out(x)
```

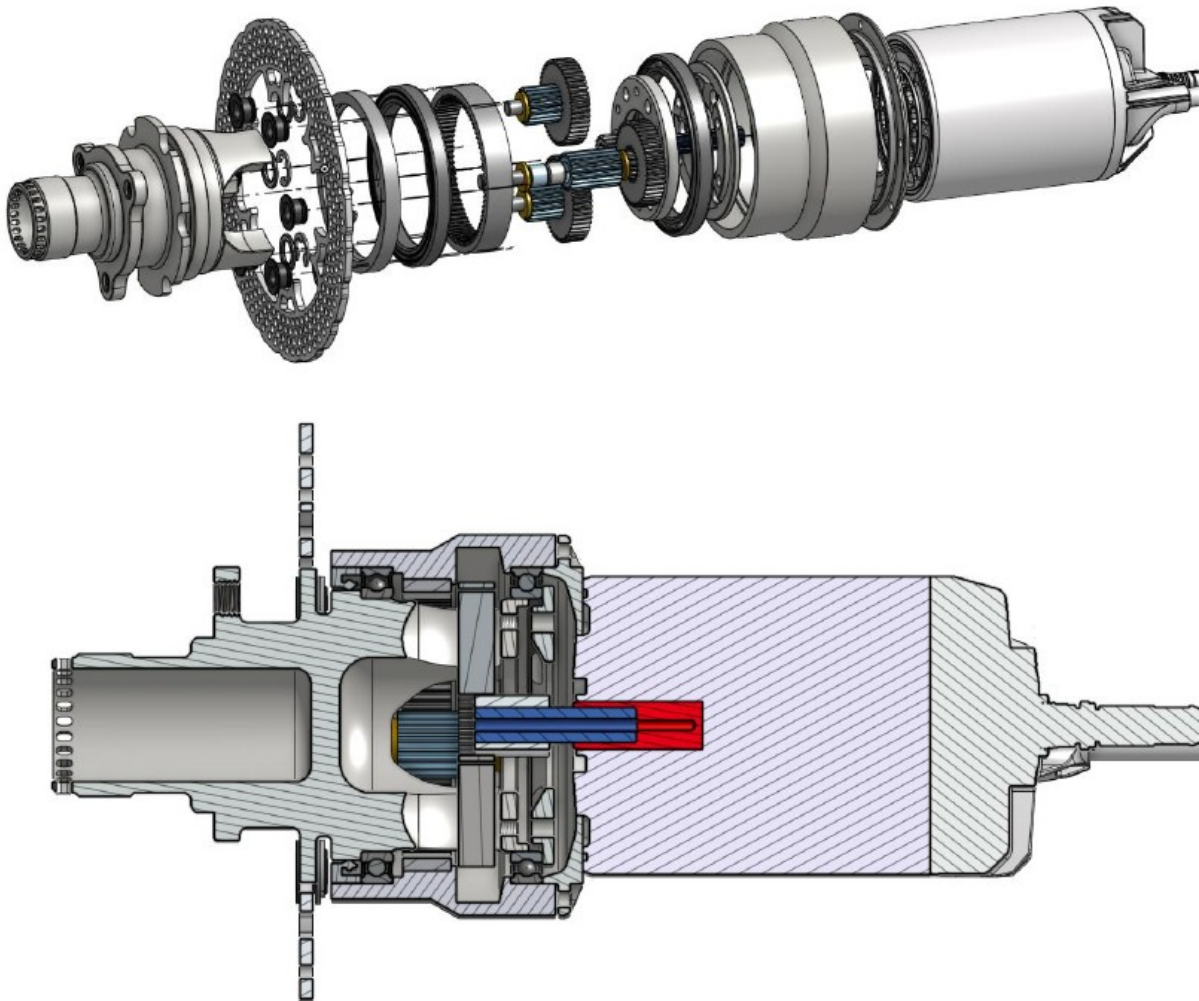
Implemented

- Cone-beam ray sampling and positional encoding
- TensorFlow network predicting local attenuation
- Discretised Beer-Lambert integration
- Residual multi-head self-attention and convolution variants
- Comparison of reconstructed slices over training

3. FORMULA STUDENT GEARBOX - ANALYSIS TO MANUFACTURE

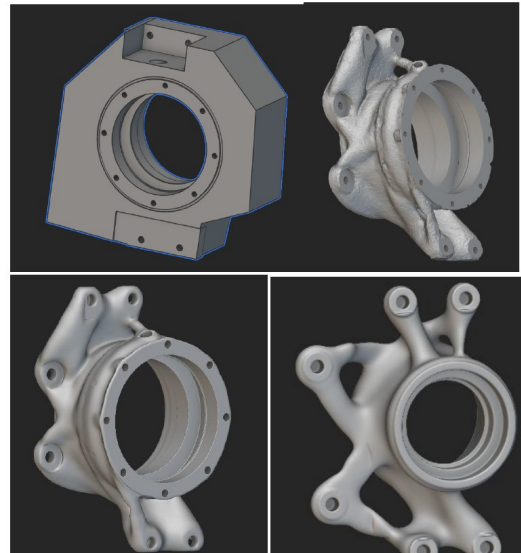
Lead mechanical transmission engineer, UC Motorsport

I was responsible for the transmission from motors to wheels: gear research, ratio selection, detailed gear geometry, bearing and geartrain analysis, CAD and drawings, supplier coordination, manufacture, processing and integration.



Exploded and section views of the complete transmission package.

3B. MANUFACTURE AND TOPOLOGY OPTIMISATION



Specific contributions

- Python searches for valid stepped-planet configurations and optimal ratio
- KISSsoft/KISSsys strength, life and load analysis
- Topology-optimised titanium housing in nTop
- Self-taught wire-EDM operation; gear set completed in under two weeks
- Secured sponsored isotropic finishing and coating

4. INDUSTRY R&D - ELECTRONICS, DIAGNOSTICS AND MACHINE BUILDING

Data Collection Limited - modular laser road-survey systems



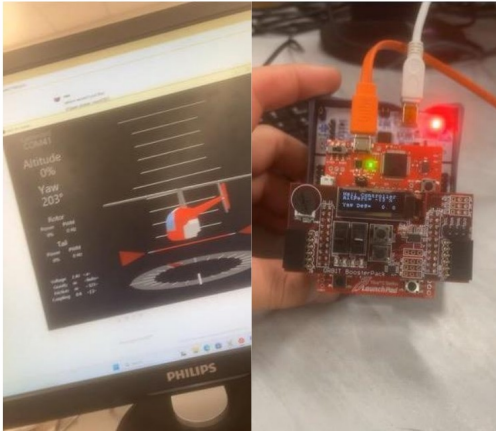
- Redesigned a daughter-board circuit in KiCad.
- Investigated jitter in a wheel-mounted high-resolution odometer that was falsely triggering data acquisition.
- Configured an NFS workflow so development boards could be programmed remotely from office PCs.

SPS Automation - small-scale injection moulder



- Co-designed and built the complete machine with one other intern.
- 20-ton hydraulic actuation, custom 316 stainless-steel barrel/ram and band heating.
- Worked through mechanical layout, hydraulics, thermal design, controls, safety, fabrication, assembly and TIG welding.

5. ADDITIONAL TECHNICAL BREADTH

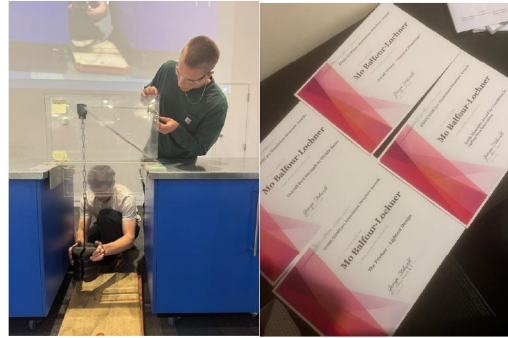


Embedded systems

C on Tiva/TM4C hardware, later extended through a multi-task FreeRTOS fitness-monitor architecture with queues, UART CLI, sensor interfaces and display control.

WORKING STYLE

The common thread across these projects is not a particular tool. It is taking responsibility for the awkward middle of a project: understanding the physics, writing the analysis, building the hardware or software, testing what does not work, and leaving behind something another person can use.



Design competition

A two-member aluminium structure designed to fail at a calculated load won all four awards among approximately 160 teams. It was the only perfect-score design and 40% lighter than the next-lightest entry.